E – 110311

Register No. | | | | | | | | | | | | | | | |

BE/BTech Degree Examination November 2025

Seventh Semester

Common to EIE, IT and Chemical Engineering

22MAO09 – OPTIMIZATION FOR ENGINEERS

(Regulations 2022)

Time: Three hours Maximum: 100 marks

Answer all Questions

Part – A (10 × 2 = 20 marks)

1. State the advantages of linear programming problem. [CO1,K2]
2. Infer slack and surplus variables. [CO1,K2]
3. Interpret Zero-one programming problem. [CO2,K2]
4. Write briefly about the Branch and Bound Technique. [CO2,K2]
5. Infer Dynamic programming. [CO3,K2]
6. Outline the characteristics of Dynamic programming. [CO3,K2]
7. Summarize the term Queue discipline. [CO4,K2]
8. Illustrate the kendal's notation for representing Queueing models. [CO4,K2]
9. Classify Kuhn-Tucker conditions in Non-linear programming. [CO5,K2]
10. Demonstrate Lagrange's function in NLPP. [CO5,K2]

Part – B (5 × 16 = 80 marks)

11 a. i) A company manufactures 2 types of printed circuits. The related data are given below: (8) [CO1,K3]

	circuit		Stock available
	A	B	
Transistor	15	10	180
Resistor	10	20	200
Capacitor	15	20	210
Profit	Rs.5	Rs.8	

Make use of graphical method, How many circuits of each type should the company produce from the stock to earn maximum profit.

ii) Construct the non negative values of x_1, x_2 and x_3 which (8) [CO1,K3]

Maximize $Z = 3x_1 + 2x_2 + 5x_3$

Subject to $x_1 + 4x_2 \leq 420$

$3x_1 + 2x_3 \leq 460$

$x_1 + 2x_2 + x_3 \leq 430$

(OR)

b. Use Big-M method to solve (16) [CO1,K3]

Maximize $Z = 4x_1 + 3x_2$
Subject to $2x_1 + x_2 \geq 10$
$-3x_1 + 2x_2 \leq 6$
$x_1 + x_2 \leq 6$
and $x_1, x_2 \geq 0$.

12. a. Identify the optimum integer solution to the following LPP: (16) [CO2,K3]

Max $Z = x_1 + x_2$
Subject to $3x_1 + 2x_2 \leq 5$
$x_2 \leq 2$
and $x_1 > 0, x_2 \geq 0$ and are integers.

(OR)

b. Solve the following mixed integer programming problem: (16) [CO2,K3]

Maximize $Z = x_1 + x_2$
Subject to the constraints
$2x_1 + 5x_2 \leq 16$
$6x_1 + 5x_2 \leq 30$
$x_2 \geq 0$, x_1, a non negative integer.

13. a. Solve the following by dynamic programming (16) [CO3,K3]

Minimize $Z = y_1 + y_2 + ... + y_n$

Subject to the constraints

$y_1, y_2 ..... y_n = b$
and $y_1, y_2, ... y_n \geq 0$.

(OR)

b. Solve the following dynamic programming approach. (16) [CO3,K3]

Max $Z = 4x_1 + 14x_2$
Subject to $2x_1 + 7x_2 \leq 21$
$7x_1 + 2x_2 \leq 21$
and $x_1, x_2 \geq 0$.

14. a. There are three typists in an office. Each typist can type an average of 6 letters per hour, if letters arrive for being typed at the rate of 15 letters per hour. (16) [CO4,K2]

1) Demonstrate in what fraction of the time all the typists will be busy?
2) Find the average number of letters waiting to be typed.
3) Demonstrate the average time a letters has to spend for waiting.
4) Extend the probability that a letter will take longer than 20 min waiting to be typed and being typed.

(OR)

b. At a port there are 6 unloading berths and 4 unloading crews. When all the berths are full, arriving ships are diverted to an over flow facility 20kms down the river. Tankers arrive according to Poisson process with a mean of 1 every 2hr. It takes for an unloading crew, on the average, 10hr to unload a tanker, the unloading time following an exponential distribution. Calculate (16) [CO4,K2]

1) how many tankers are at the port on the average.

2) how long does a tanker spend at the port on the average?

3) the average arrival rate at the overflow facility.

15. a. Develop the optimum solution of the following constrained multivariable problem. (16) [CO5,K3]

Minimize $Z=x_1^2+(x_2+1)^2+(x_3-1)^2$

Subject to the constraints

$$x_1+5x_2-3x_3=6$$
and $x_1,x_2,x_3\geq 0$

(OR)

b. Construct the optimum value of the objective function when separately subject to the following constraints: (16) [CO5,K3]

Maximize $z=10x_1-x_1^2+10x_2-x^2$,

Subject to

$$x_1+x_2\leq 14$$
$$-x_1+x_2\leq 6$$
and $x_1,x_2\geq 0$.

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	–	29	71	–	–	–

Kongu Engineering College, Perundurai Erode

B.E / B.Tech Examinations Nov. 2025

VII Semester

Common to EIE, IT and Chemical Engineering

22MA009 - OPTIMIZATION FOR ENGINEERS

[Scheme of the Valuation]

Part A (10 x 1 = 10 marks)

1. Advantages of LPP (Any Two Advantages)

1. It is a procedure provides an insight and perspective to the problem environment. This generally results in clear picture of the true problem. (1)
2. It makes a scientific and mathematical analysis of the problem situations (1)
3. It gives an opportunity to the decision maker to formulate his strategies consistent with the constraints and the objectives

2. For the general LPP

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_j \ (j = 1, 2, 3, \ldots k)$$

The non-negative variable s_i which are introduced to convert the inequalities to equalities

$\sum_{j=1}^{n} a_{ij} x_j + s_i = b_j$ are called slack variables (1)

(ii)ly The non-negative variables s_j are introduced to convert the inequalities in $\sum_{j=1}^{n} a_{ij} x_j \geq b_j$ are called Surplus variables. (1)

3. If all the variables in the optimum solution of Integer programming problem are allowed to take the values either zero or one (0 or 1) type decision, then the problem is called Zero one programming problem (2)

4) Branch and bound technique starts with continuous optimum but systematically partitions the solution space into two problems that eliminates parts that contain no feasible integer solution. (2) (2)

5) Dynamic programme is a mathematical Technique of optimization using multistage decision process. in which a sequence of interrelated decisions has to be made. (2)

6) 1) The problems are divided into stages, with a policy decision required at each stage. (1)

2) Each stage has n number of states associated with it. (2)

3). the effect of policy decision at each stage is to transform the current state to next.

7) The quaeuediscipline is the rule determining the formation of queue, the number of customer's behaviour while waiting, and the manner inwhich they are choosen for service. (2)

8) Any queueing model is always represented by (a/b/c):(d/e) where

a → the number of arrivals
b → service time
c → No of servers
d → capacity of the system
e → queue discipline (2)

9) Kuhn-Tucker conditions are

i) $\frac{\partial f(x)}{\partial x_i} - \sum_{i=1}^{m} \lambda_i \frac{\partial g_i(x)}{\partial x_i} = 0, \quad i = 1, 2, \ldots n$

ii) $\lambda_i g_i(x) = 0$

iii) $g_i(x) \leq 0$

iv) $\lambda_i \geq 0$ (2)

10) Lagrangean function is denoted by $L(x_j, \lambda)$ (3)
and can be expressed as

$$L(x_j, \lambda) = f(x_j) - \lambda g(x_j) \qquad j = 1, 2, 3 \ldots$$ (2)

PART B

(4)

11 a) i) The complete formulation of LPP is

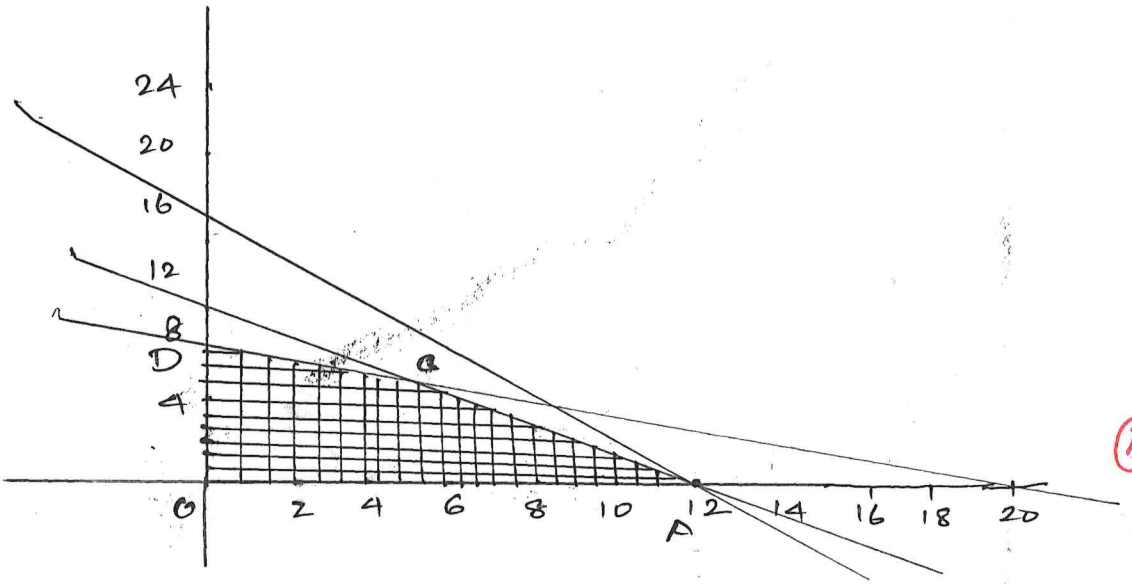
$$\max z = 5x_1 + 8x_2$$

s.t.c $15x_1 + 10x_2 \leq 180$

$10x_1 + 20x_2 \leq 200$

$15x_1 + 20x_2 \leq 210$, $x_1, x_2 \geq 0$ (2)

Using graphical method, we can get the vertices from the constraints, O (0,0), A (12,0) B (10,3), C (2,9) and D (0,10)



(4)

The values of Z of these vertices

O (0,0) = 0 | C (2,9) = 82

A (12,0) = 60 | D (0,10) = 80

B (10,3) = 74

$\Rightarrow$ Max Z = 82, $x_1 = 2$, $x_2 = 9$ (2)

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a) ii) The standard form of given LPP is

$$\text{Max } z = 3x_1 + 2x_2 + 5x_3 + 0s_1 + 0s_2 + 0s_3$$

s.t.c, $x_1 + 4x_2 + 0x_3 + s_1 + 0s_2 + 0s_3 = 420$

$3x_1 + 0x_2 + 2x_3 + 0s_1 + s_2 + 0s_3 = 460$

$x_1 + 2x_2 + x_3 + 0s_1 + 0s_2 + s_3 = 430$ (2)

Simplex table is. $x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$

			3	2	5	0	0	0	
C_B	Y_B	X_B	x_1	x_2	x_3	s_1	s_2	s_3	Ratio
0	s_1	420	1	4	0	1	0	0	
0	s_2	460	3	0	(2)	0	1	0	$\frac{460}{2} = 230$ (X)
0	s_3	430	1	2	1	0	0	1	
	$z_j - c_j$	0	-3	-2	-5	0	0	0	(3)

Using Simplex method, we can get (5)

			3	2	5	0	0	0	
C_B	Y_B	X_B	x_1	x_2	x_3	s_1	s_2	s_3	Ratio
0	s_1	20	2	0	0	1	1	-2	
5	x_3	230	3/2	0	1	0	1/2	0	
2	x_2	100	-1/2	1	0	0	-1/4	1/2	
$Z_j - C_j$		1350	4	0	0	0	2	1	

(2)

Since all $Z_j - C_j \geq 0$ ∴ Optimal solution is

Max $Z = 1350$, $x_1 = 0$, $x_2 = 100$, $x_3 = 230$ (1)

11. b) The standard form of LPP is converted from the given LPP as

$$\text{Max } Z = 4x_1 + 3x_2 + 0s_1 + 0s_2 + 0s_3 - MR_1 - MR_2$$

Stc, $2x_1 + x_2 - s_1 + 0s_2 + 0s_3 + R_1 = 10$

$-3x_1 + 2x_2 + 0s_1 + s_2 + 0s_3 = 6$

$x_1 + x_2 + 0s_1 + 0s_2 - s_3 = 6$

$x_1, x_2, s_1, s_2, s_3, R_1, R_2 \geq 0$ (4)

Corresponding Simplex table is

			4	3	0	0	0	-M	-M	
C_B	Y_B	X_B	x_1	x_2	s_1	s_2	s_3	R_1	R_2	Ratio
-M	R_1	10	(2)	1	-1	0	0	1	0	$\frac{10}{2} = 5$
0	s_2	6	-3	2	0	1	0	0	0	
-M	R_2	6	1	1	0	0	-1	1	1	$\frac{6}{1} = 6$
$Z_j - C_j = -16M$			-3M+4	-2M+3	M	0	M	0	0	(4)
-4	x_1	5	1	1/2	-1/2	0	0		0	10
0	s_2	21	0	7/2	-3/2	1	0		0	42/7
-M	R_2	1	1	(1/2)	1/2	0	-1		1	2
$Z_j - C_j$		-M-20	0	-M/2+1/2	-M/2+2	0	M		0	(4)

Since there are some $z_j - c_j < 0$ so the current feasible solution is not optimal. Using the same process, we can get the simplex table as.

			4	3	0	0	0	
C_B	Y_B	X_B	x_1	x_2	s_1	s_2	s_3	Ratio
-4	x_1	-4	1	0	-1	0	1	
0	s_2	14	0	0	-5	1	7	
-3	x_2	-2	0	1	1	0	-2	
		22	0	0	0	0	2	

∴ Optimal soln is Max $Z = 22$, $x_1 = -4$ & $x_2 = -2$ (4)

12a The standard form of LPP is

$$\text{Max } Z = x_1 + x_2 + 0x_3 + 0x_4$$

$$\text{s.t.} \quad 3x_1 + 2x_2 + x_3 + 0x_4 = 5$$

$$0x_1 + x_2 + 0x_3 + x_4 = 2$$

$$x_1, x_2, x_3, x_4 \geq 0$$

The initial basic feasible solution is $x_3 = 5$ & $x_4 = 2$, ($x_1 = 0$, $x_2 = 0$ are non-basic feasible solutions) (3)

Initial Approximation

			1	1	0	0	
C_B	Y_B	X_B	x_1	x_2	x_3	x_4	θ
0	x_3	5	(3)	2	1	0	5/3
0	x_4	2	0	1	0	1	—
$z_j - c_j$		0	-1 ↑	-1	0	0	

Since some of $z_j - c_j < 0$, the current basic feasible solution is not optimal, (4)

C_B	Y_B	X_B	x_1	x_2	x_3	x_4	θ
1	x_1	5/3	1	2/3	1/3	0	3/2
0	x_4	2	0	(1)	0	0	2 ✱
$z_j - c_j$		5/3	0	-1/3	1/3	0	
1	x_1	1/3	1	0	1/3	-2/3	
1	x_2	2	0	1	0	1	(4)
$z_j - c_j$		7/3	0	0	1/2	1/2	

To obtain optimum integer solution, we have (7) to add fractional constraints row in the optimal simplex table, therefore

			1	1	0	0	0	
C_B	Y_B	X_B	x_1	x_2	x_3	x_4	s_1	θ
1	x_1	$1/3$	1	0	$1/3$	$-2/3$	0	
1	x_2	2	0	1	0	1	0	
0	s_1	$-1/3$	0	0	$-1/3$	$-1/3$	1	(2)
		$7/3$	0	0	$1/3$	$1/3$	0	

Here solution is optimal but infeasible, so drop s_1 and introduce x_3, we get-

			1	1	0	0	0	
C_B	Y_B	X_B	x_1	x_2	x_3	x_4	s_1	θ
1	x_1	1	1	0	0	-1	1	
1	x_2	2	0	1	0	1	1	
0	x_3	1	0	0	1	1	-3	
→		2	0	0	0	0	1	

Since all $z_j - c_j \geq 0$ and all $X_{Bj} \geq 0$. Then the current solution is feasible and optimal and integer (3)

$\therefore$ Max $Z = 2$, $x_1 = 0$, $x_2 = 2$

12 b) Standard form of LPP is Max $Z = x_1 + x_2$

s.t, $2x_1 + 5x_2 + 1x_3 + 0x_4 = 16$

$6x_1 + 5x_2 + 0x_3 + x_4 = 30$

$x_1, x_2, x_3, x_4 \geq 0$ (3)

Here $x_3 = 16$, $x_4 = 30 \rightarrow$ initial Basic feasible solⁿ

& $x_1 = 0, x_2 = 0$, are Non basic feo. sol

			1	1	0	0	
C_B	Y_B	X_B	x_1	x_2	x_3	x_4	θ
0	x_3	16	2	5	1	0	8
0	x_4	30	(6) ↑	5	0	1	5
$z_j - c_j$		0	-1	-1	0	0	(2)
0	x_3	6	0	(10/3)	1	$-1/3$	9/5
1	x_1	5	1	5/6 ↑	0	1/6	6
$z_j - c_j$		5	0	$-1/6$	0	1/6	(2)

Add the Gomorian constraints to the bottom of above Simplex table, we have (8)

			1	1	0	0	0	
C_B	Y_B	X_B	x_1	x_2	x_3	x_4	s_1	
1	x_2	$9/5$	0	1	$3/10$	$-1/10$	0	
1	x_1	$7/2$	1	0	$-1/4$	$1/4$	0	
0	s_1	$-1/2$	0	0	$-1/4$	$-1/4$	1	
$Z_j - C_j$		$53/10$	0	0	$1/20$	$3/20$	0	(3)
1	x_2	$6/5$	0	1	0	$-2/5$	$6/5$	
1	x_1	4	1	0	0	$1/2$	-1	
0	x_3	2	0	0	1	1	-4	
$Z_j - C_j$		$26/5$	0	0	0	$1/10$	$1/5$	(4)

by droping s_1 and introduced x_3, therefore

$\text{Max } Z = 26/5$, $x_1 = 4$ $x_2 = 6/5$ (2)

13a) To develop the recursive equation, $f_n(b)$ be the minimum attainable sum $y_1 + y_2 + \cdots + y_n$ (2)

For $n = 1 \Rightarrow f_1(b) = \min_{y_1 = b} \{y_1\} = b$ (trivial case) (2)

For $n = 2 \Rightarrow f_2(b) = \min_{0 \le x \le b} \{y_1 + y_2\} = \min_{0 \le x \le b} \{x + \frac{b}{x}\}$ (2)

For $n = 3 \Rightarrow f_3(b) = \min_{0 \le x \le b} \{x + f_2(\frac{b}{x})\}$ (2)

Ily For n, $f_n(b) = \min_{0 \le x \le b} \{x + f_{n-1}(b/x)\}$

The optimal policy of the above cases are

$f_2(b) = 2\sqrt{b} = 2b^{1/2}$; $f_3(b) = 3b^{1/3}$ (4)

$\Rightarrow f_n(b) = n\, b^{1/n}$

So $f_{m+1}(b) = \min_{0 \le x \le b} \{x + m(b/x)^{1/m}\}$

∴ By mathematical induction, the Optimal policy is $(b^{1/n}, b^{1/n} \ldots b^{1/n})$ and

$f_m(b) = m\, b^{1/m}$ (4)

$\Rightarrow f_n(b) = n\, b^{1/n}$

13 b) Given problem consists of two decision variables and two constraints, so the problem has two stages. (9)

Let (B_{1j}, B_{2j}) be states of the system. and $f_j^*(B_{1j}, B_{2j})$ be the optimal values. Using backward computational procedure, (2)

$$f_2(B_{12}, B_{22}) = \underset{\substack{0 \le 7x_2 \le B_{12} \\ 0 \le 2x_2 \le B_{22}}}{\text{Max}} \{14x_2\}$$

$$= 14 \text{ Min} \left\{ \frac{B_{12}}{7}, \frac{B_{22}}{2} \right\} \quad (4)$$

Also $f_1(B_{11}, B_{21}) = \underset{\substack{0 \le x_1 \le \frac{B_{11}}{2} \\ 0 \le x_1 \le \frac{B_{21}}{7}}}{\text{Max}} \left\{ 4x_1 + 14 \text{ Min} \left(\frac{B_{11} - 2x_1}{7}, \frac{B_{21} - 7x_1}{2} \right) \right\}$ (4)

Proceeding with the first stage values,

$$f_1(B_{11}, B_{21}) = \text{Max} \begin{cases} 42 & 0 \le x_1 \le 7/3 \\ 147 - 45x_1 & 7/3 \le x_1 \le 3 \end{cases} \quad (4)$$

∴ There are infinitely many solutions

Let $x_1 = \lambda \Rightarrow x_2 = \frac{21 - 2\lambda}{2}$ where $0 \le \lambda \le 7/3$ and

$\text{Max } z = 42$ (2)

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14 a) The given problem is $(M/M/s) : (\infty/FIFO)$ model.

$\lambda = 15; \ \mu = 6, \ s = 3$

1) P(all typists are busy) $= \dfrac{\left(\frac{\lambda}{\mu}\right)^3 \cdot P_0}{3!\left(1 - \frac{\lambda}{3\mu}\right)} = \dfrac{1}{22.25} = 0.044$ (4)

2) $E(N_q) = \dfrac{1}{s \cdot s!} \dfrac{\left(\frac{\lambda}{\mu}\right)^{n+1} P_0}{\left(1 - \frac{\lambda}{\mu s}\right)^2} = 3.5078$ (4)

(4)

3) $E(W) = \dfrac{1}{\lambda} E(N) =$ 24 minutes approximately (4)

4) $P(W > 1/3) = 0.4616$

14 b) Here $\lambda = \frac{1}{2}$ $\mu = 1/10$ $s = 4$, $k = 6$ (10)

1) $P_0 = \left\{ \sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n + \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \sum_{n=s}^{k} \left(\frac{\lambda}{\mu s}\right)^{n-s} \right\}^{-1}$

$\Rightarrow P_0 = 0.0072$

$E(N_q) = \left(\frac{\lambda}{\mu s}\right) \cdot \left(s! \left(1 - \frac{\lambda}{\mu s}\right)^2 \right)^{-1} \cdot P_0 \left(\frac{\lambda}{\mu}\right)^s$

$\rho = 0.8203$

$\Rightarrow E(N_0) = 4.8203$

$E(N) = E(N_0) - \{4P_0 + 3P_1 + 2P_2 + P_3\} = 4.3535$

2) $E(W) = \frac{1}{\lambda'}$, $\lambda' = 0.3533$

$= 12.32$ hr

3) $P(N = 6) = \frac{1}{s! \cdot s^{n-1}} \left(\frac{\lambda}{\mu}\right)^n \cdot P_0 = 0.2930$

$\Rightarrow$ Average arrival rate at the over flow facility $= \frac{1}{2}(0.2930)$

$= 0.586$ per hour

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15 a) Since the given problem has 3 variables and one equality constraint, any one of the variable can be removed from z

$\therefore x_3 = \frac{x_1 + 5x_2 - 6}{3}$ and substituting we get

$z = f(x) = x_1^2 + (x_2 + 1)^2 + \frac{1}{9}(x_1 + 5x_2 - 9)^2$

The neccesary Condition for minimum of Z is

$\nabla z = \nabla f(x) = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}\right] = 0$

$= \left[2x_1 + \frac{2}{9}(x_1 + 5x_2 - 9), \; 2(x_2 + 1) + \frac{10}{9}(x_1 + 5x_2 - 9) \right]$

Solving $\frac{\partial f}{\partial x_1} = 0$ & $\frac{\partial f}{\partial x_2} = 0 \Rightarrow x_1 = 2/5$ & $x_2 = 1$

$$\therefore W(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 z}{\partial x_1^2} & \frac{\partial^2 z}{\partial x_1 \partial x_2} \\ \frac{\partial^2 z}{\partial x_2 \partial x_1} & \frac{\partial^2 z}{\partial x_2^2} \end{bmatrix} \quad (11)$$

$$= \begin{pmatrix} 20/9 & 10/9 \\ 10/9 & 20/9 \end{pmatrix}$$

Min $z = 28/5$; $x_1 = 2/5$; $x_2 = 1$; $x_3 = -1/5$ (4)

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15 b) Here the constraints are

$$g_1(x) = x_1 + x_2 + s_1^2 - 14 = 0$$

$$g_2(x) = -x_1 + x_2 + s_2^2 - 6 = 0 \quad (2)$$

The Lagrangian function is

$$L(x_1, s, \lambda) = (10x_1 - x_1^2 + 10x_2 - x_2^2) - \lambda_1(x_1 + x_2 + s_1^2 - 14) - \lambda_2(-x_1 + x_2 + s_2^2 - 6) \quad (2)$$

The Kuhn-Tucker necessary conditions for maximization problem are

$$\frac{\partial L}{\partial x_1} = 0 \Rightarrow 10 - 2x_1 - \lambda_1 + \lambda_2 = 0 \longrightarrow (1)$$

$$\frac{\partial L}{\partial x_2} = 0 \Rightarrow 10 - 2x_1 - \lambda_1 - \lambda_2 = 0 \longrightarrow (2) \quad (2)$$

$$\frac{\partial L}{\partial \lambda_1} = 0 \Rightarrow -(x_1 + x_2 + s_1^2 - 14) = 0 \longrightarrow (3) \quad (2)$$

$$\frac{\partial L}{\partial \lambda_2} = 0 \Rightarrow -(-x_1 + x_2 + x_2^2 - 6) = 0 \longrightarrow (4) \quad (2)$$

$$\frac{\partial L}{\partial s_1} = 0 \Rightarrow -2\lambda_1 s_2 = 0 \longrightarrow (5)$$

$$\frac{\partial L}{\partial s_2} = 0 \Rightarrow -2\lambda_2 s_2 = 0 \longrightarrow (6) \quad (2)$$

Since the unconstrained solution (~~are~~ i.e $\lambda_1 = 0, \lambda_2 = 0$ is obtained from first 4 equations. We sel-

(12)

$\bar{x}_1 = 5$; $\bar{x}_2 = 5$, $s_1^2 = 4$, $s_2^2 = 6$

and Max $z = 50$

Now $H = \begin{bmatrix} \frac{\partial^2 z}{\partial x_1^2} & \frac{\partial^2 z}{\partial x_1 \partial x_2} \\ \frac{\partial^2 z}{\partial x_2 \partial x_1} & \frac{\partial^2 z}{\partial x_2^2} \end{bmatrix}$

$= \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$ (2)

$\det A_1 = |A_1| = 2$ $\det A_2 = |A_2| = 4$

Since the signs of principal Minor of H are alternating matrix H is negative definite and the point $(4,4)$ gives the local maximu of the objective function z (2)